PROJECT DESCRIPTION

FieldShift Ethiopia – Clear Planting & Residual Moisture Decisions for Rain-fed Smallholder Farmers Using NASA Earth Data

1. Executive Summary

FieldShift Ethiopia is a farmer-centered decision-support system that transforms multiple open NASA Earth observation datasets into simple, actionable advice for smallholder farmers in Ethiopia. The system is built around the real decisions that determine whether a farming season succeeds or fails: when to plant, how to respond to sudden dry spells, and whether residual soil moisture after the main rains can support a short second crop.

Unlike generic climate dashboards that display maps and technical indicators, FieldShift Ethiopia answers three practical questions in plain language (English and Amharic):

- Is the soil wet enough to plant this week in my woreda?
- Is a dangerous dry spell (flash drought) starting after planting?
- After the main rains end, is there still enough moisture for chickpea, lentil, or short vegetables?

The solution is deliberately designed for dual delivery: a lightweight mobile and web application for smartphone users, and an SMS service for the majority of rural farmers who still use basic phones. Both channels are intended to work in partnership with the Ethiopian Ministry of Agriculture and the national network of Development Agents, so that the advice reaches farmers through trusted institutional channels.

2. Problem Statement – Why This Matters in Ethiopia

Ethiopia's agriculture remains overwhelmingly rain-fed. More than 80% of the population depends on smallholder farming, and the majority of cultivated land receives no irrigation. The two main seasons — Belg (short rains) and Meher (main rains) — are increasingly unreliable. Farmers frequently face four costly problems:

- **Planting into dry soil:** Many farmers plant after the first showers only to watch seeds fail to germinate or seedlings die when rain stops.
- **Sudden dry spells (flash droughts):** Even after good early rains, 2–3 weeks without rain during critical growth stages can destroy the crop.
- **Missed residual moisture opportunities:** After the main rains end, significant moisture often remains in the root zone. This moisture can support short-cycle legumes (chickpea, lentil, grass pea) or vegetables, yet most farmers do not receive clear information about this window.
- **Rain at harvest:** Unexpected late rains can spoil teff, wheat, and barley that are ready for harvest.

Existing climate information services are often too coarse (regional or national averages), arrive too late, or present data in forms that farmers and even many Development Agents cannot easily turn into a decision. The result is repeated loss of seed, labor, and potential income.

3. Our Approach – Focused, Honest, and Actionable

FieldShift Ethiopia does not try to solve every agricultural problem with satellite data. We deliberately focus on the water-related decisions where NASA Earth observation data is strongest and where the cost of wrong decisions is highest for smallholders.

We reject the common pattern of showing farmers complex maps, anomaly charts, or raw soil moisture percentages. Instead we built three focused decision engines that fuse multiple NASA datasets and output only clear recommendations.

4. The Three Decision Engines

4.1 Planting Readiness Engine

This engine answers the question: "Is it safe to plant now in my woreda?" It combines root-zone and surface soil moisture with recent rainfall performance. Research in the Upper Blue Nile Basin and other Ethiopian highlands has shown that soil moisture at planting time has a strong influence on final yield. By focusing on root-zone moisture (not only surface wetness), the engine reduces the risk of farmers planting into soil that looks wet on top but is dry below.

4.2 Flash Drought Watch Engine

Flash droughts — rapid onset dry spells — are becoming more common and are especially damaging because they occur after farmers have already invested seed and labor. This engine monitors the combination of rapidly declining soil moisture, rising evaporative stress, and rainfall deficits. It is designed to give an early warning so that farmers and Development

Agents can take protective actions while there is still time.

4.3 Residual Moisture Opportunity Engine

After the main Meher rains decline, substantial moisture often remains in the soil profile, particularly in the western, central, and southwestern highlands. This residual moisture can support short-duration legumes and vegetables. Most current tools ignore this window. Our engine specifically evaluates remaining root-zone moisture against the water requirements of common short-cycle crops and tells the farmer whether a second planting is realistic.

Summary of the Three Engines:

Engine	Core Farmer Question	Primary NASA Inputs	Output to Farmer
Planting Readiness	Is the soil ready to plant this week?	SMAP L4 root-zone, NASA-USDA Soil Moisture, GPM IMERG	Plant Now / Plant with Caution / Wait
Flash Drought Watch	Is a sudden dry spell developing?	ECOSTRESS, SMAP, GPM, MODIS ESI & LST	No Risk / Watch Closely / High Risk
Residual Moisture Opportunity	Can I plant a short crop after the rains?	SMAP root-zone, FLDAS, GLDAS, GPM	Good Opportunity / Limited / Not Recommended

5. Backend Architecture – How We Integrate and Interpret NASA Data

A key differentiator of FieldShift Ethiopia is that we do not simply display satellite layers. We implement a clear three-layer interpretation pipeline.

5.1 Data Ingestion Layer

The system is designed to ingest open NASA products through Google Earth Engine, NASA Earthdata, or pre-processed regional archives. For any selected woreda or GPS coordinate we extract time series and near-real-time values for:

- SMAP L4 root-zone soil moisture (and surface soil moisture)
- NASA-USDA Global Soil Moisture products
- GPM IMERG precipitation (early and final runs)
- ECOSTRESS land surface temperature and evaporative stress
- MODIS-based Evaporative Stress Index (ESI) and land surface temperature
- FLDAS and GLDAS land surface model outputs (soil moisture layers, anomalies)
- Supporting climate context from NASA POWER (temperature, humidity, solar radiation, historical normals)

5.2 Interpretation / Fusion Layer

Raw values are converted into decision-relevant indicators using rules tailored to Ethiopian rain-fed systems:

- **Moisture Adequacy:** Root-zone moisture is compared against crop-specific and agro-ecological thresholds rather than using a single national cutoff.
- **Recent Rainfall Support:** Short-term rainfall accumulation and anomalies from GPM IMERG are used to confirm whether moisture is being replenished or is declining.
- **Drying Rate & Stress:** The combination of falling soil moisture and rising ECOSTRESS / ESI values signals the onset of flash drought conditions.
- **Residual Moisture Potential:** After the main rains decline, remaining root-zone moisture is evaluated against the estimated water requirement of short-cycle legumes and vegetables commonly grown in Ethiopia.

These rules are intentionally transparent and can be adjusted with local agronomic knowledge as the system moves from prototype to operational use.

5.3 Translation Layer (Farmer-Facing Output)

The final and most important step is translation. Technical outputs (anomalies, volumetric moisture, stress indices) are converted into short, clear statements and traffic-light recommendations in both English and Amharic. Farmers never see the underlying satellite numbers — they only see the decision and a short explanation.

6. Delivery Channels – Reaching Farmers Where They Are

Technology only creates impact if it reaches the user. FieldShift Ethiopia is therefore built as a dual-channel system from the beginning.

Mobile and Web Application

A lightweight, low-bandwidth interface that allows users to select region and woreda, choose a crop, and instantly receive the three engine recommendations as color-coded cards. The interface supports English and Amharic and is designed so that Development Agents can also use it during farmer meetings.

SMS Service

Farmers without smartphones can send a short code (for example “PLANT” followed by their woreda name) and receive the core recommendation by text message. This channel is essential for equity and scale in rural Ethiopia.

Partnership with Government Extension System

Sustainable impact requires institutional ownership. The system is designed to complement, not replace, the existing extension structure. Development Agents can access the richer app interface and then communicate simplified advice to farmer groups. Alignment with the Ministry of Agriculture and regional bureaus is a core part of the long-term vision.

7. Why This Approach Is Competitive

Many hackathon projects display NASA data. Fewer projects show a clear chain from satellite observation → interpreted decision → farmer action. FieldShift Ethiopia is competitive because:

- It focuses on high-cost, water-related decisions that NASA data can actually support well.
- It uses a combination of root-zone moisture (SMAP L4) and thermal/evaporative stress (ECOSTRESS) that is still under-utilized in simple farmer tools.
- It explicitly addresses residual moisture — an opportunity that is scientifically documented in Ethiopia but poorly served by existing advisory services.
- It includes a realistic delivery strategy (App + SMS + government extension) instead of assuming every farmer has a smartphone and high digital literacy.
- It is honest about the limits of satellite data (we do not claim to solve pest, disease, or market problems with NASA observations).

8. Current Prototype Status

A working interactive HTML prototype has been developed. It currently allows users to:

- Select Ethiopian region and woreda from dropdown lists
- Choose from a wide range of crops (teff, maize, wheat, barley, sorghum, chickpea, lentil, etc.)
- Switch between English and Amharic
- Receive color-coded advice from the three decision engines

The current prototype uses intelligent regional simulation to demonstrate the user experience and decision logic. The architecture is deliberately prepared for connection to live Google Earth Engine or NASA data services in the next development stage.

9. Expected Impact and Path to Scale

If the system is refined with real data feeds and adopted through the extension system, it can contribute to:

- Fewer cases of seed and labor lost to planting into insufficient moisture
- Earlier awareness of flash-drought conditions so that farmers and local officials can respond
- Increased utilization of residual moisture for nutritious short-cycle legumes
- Stronger, data-supported conversations between Development Agents and farmers

Even modest improvements in planting success rates and second-crop adoption would have meaningful effects on household food security across Ethiopia’s rain-fed farming areas.

10. Conclusion

FieldShift Ethiopia starts from the daily decisions of Ethiopian smallholder farmers and works backward to the NASA datasets that can usefully inform those decisions. By combining root-zone soil moisture, rainfall, and evaporative stress information into three clear engines — Planting Readiness, Flash Drought Watch, and Residual Moisture Opportunity — and by delivering the results through both smartphone and SMS channels in partnership with government extension services, the project offers a practical and scalable way to turn open Earth observation data into better farming decisions.

We believe this focused, honest, and delivery-oriented approach is both useful for real farmers and competitive within the Field Shift challenge.

FieldShift Ethiopia | Clear decisions from NASA Earth Data
NASA Space Apps Challenge 2026 • Challenge: Field Shift – Adapting Farms with NASA Data